

9a)

$$\frac{X(s)}{F(s)} = \frac{7}{s^2 + 5s + 10}$$

$$X(s^2 + 5s + 10) = F \cdot 7$$

$$s^2 X(s) + 5s X(s) + 10 X(s) = 7 F(s)$$

$$\ddot{x} + 5\dot{x} + 10x = 7f(t)$$

9b)

$$\frac{X(s)}{F(s)} = \frac{15}{(s+10)(s+11)} = \frac{15}{s^2 + 21s + 110}$$

$$s^2 X(s) + 21s X(s) + 110 X(s) = 15 F(s)$$

$$\ddot{x} + 21\dot{x} + 110x = 15f(t)$$

9c)

$$\frac{X(s)}{F(s)} = \frac{s+3}{s^3 + 11s^2 + 12s + 18}$$

$$s^3 X(s) + 11s^2 X(s) + 12s X(s) + 18 X(s) = s F(s) + 3 F(s)$$

$$\ddot{x} + 11\ddot{x} + 12\dot{x} + 18x = f'(t) + 3f(t)$$